

Paul's Online Notes

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You appear to be on a device with a "narrow" screen width (*i.e.* you are probably on a mobile phone). Due to the nature of the mathematics on this site it is best viewed in landscape mode. If your device is not in landscape mode many of the equations will run off the side of your device (you should be able to scroll/swipe to see them) and some of the menu items will be cut off due to the narrow screen width.

Section 12.11 : Velocity And Acceleration

In this section we need to take a look at the velocity and acceleration of a moving object.

From Calculus I we know that given the position function of an object that the velocity of the object is the first derivative of the position function and the acceleration of the object is the second derivative of the position function.

So, given this it shouldn't be too surprising that if the position function of an object is given by the vector function $\vec{r}(t)$ then the velocity and acceleration of the object is given by,

$$\vec{v}(t) = \vec{r}'(t) \qquad \vec{a}(t) = \vec{r}''(t)$$

Notice that the velocity and acceleration are also going to be vectors as well.

In the study of the motion of objects the acceleration is often broken up into a **tangential component**, a_T , and a **normal component**, a_N . The tangential component is the part of the acceleration that is tangential to the curve and the normal component is the part of the acceleration that is normal (or orthogonal) to the curve. If we do this we can write the acceleration as,

$$\vec{a} = a_T \vec{T} + a_N \vec{N}$$

where $\vec{T}$ and $\vec{N}$ are the unit tangent and unit normal for the position function.

If we define $v = \|\vec{v}(t)\|$ then the tangential and normal components of the acceleration are given by,

$$a_T = v' = \frac{\vec{r}'(t) \cdot \vec{r}''(t)}{\|\vec{r}'(t)\|} \quad a_N = \kappa v^2 = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|}$$

where κ is the **curvature** for the position function.

There are two formulas to use here for each component of the acceleration and while the second formula may seem overly complicated it is often the easier of the two. In the tangential component, v , may be messy and computing the derivative may be unpleasant. In the normal component we will already be computing both of these quantities in order to get the curvature and so the second formula in this case is definitely the easier of the two.

Let's take a quick look at a couple of examples.

Example 1 If the acceleration of an object is given by $\vec{a} = \vec{i} + 2\vec{j} + 6t\vec{k}$ find the object's velocity and position functions given that the initial velocity is $\vec{v}(0) = \vec{j} - \vec{k}$ and the initial position is $\vec{r}(0) = \vec{i} - 2\vec{j} + 3\vec{k}$.

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We'll first get the velocity. To do this all (well almost all) we need to do is integrate the acceleration.

$$\begin{aligned} \vec{v}(t) &= \int \vec{a}(t) dt \\ &= \int \vec{i} + 2\vec{j} + 6t\vec{k} dt \\ &= t\vec{i} + 2t\vec{j} + 3t^2\vec{k} + \vec{c} \end{aligned}$$

To completely get the velocity we will need to determine the "constant" of integration. We can use the initial velocity to get this.

$$\vec{j} - \vec{k} = \vec{v}(0) = \vec{c}$$

The velocity of the object is then,

$$\begin{aligned} \vec{v}(t) &= t\vec{i} + 2t\vec{j} + 3t^2\vec{k} + \vec{j} - \vec{k} \\ &= t\vec{i} + (2t + 1)\vec{j} + (3t^2 - 1)\vec{k} \end{aligned}$$

We will find the position function by integrating the velocity function.

$$\begin{aligned}
 \vec{r}(t) &= \int \vec{v}(t) dt \\
 &= \int t \vec{i} + (2t + 1) \vec{j} + (3t^2 - 1) \vec{k} dt \\
 &= \frac{1}{2}t^2 \vec{i} + (t^2 + t) \vec{j} + (t^3 - t) \vec{k} + \vec{c}
 \end{aligned}$$

Using the initial position gives us,

$$\vec{i} - 2\vec{j} + 3\vec{k} = \vec{r}(0) = \vec{c}$$

So, the position function is,

$$\vec{r}(t) = \left(\frac{1}{2}t^2 + 1 \right) \vec{i} + (t^2 + t - 2) \vec{j} + (t^3 - t + 3) \vec{k}$$

Example 2 For the object in the previous example determine the tangential and normal components of the acceleration.

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There really isn't much to do here other than plug into the formulas. To do this we'll need to notice that,

$$\begin{aligned}
 \vec{r}'(t) &= t \vec{i} + (2t + 1) \vec{j} + (3t^2 - 1) \vec{k} \\
 \vec{r}''(t) &= \vec{i} + 2\vec{j} + 6t\vec{k}
 \end{aligned}$$

Let's first compute the dot product and cross product that we'll need for the formulas.

$$\vec{r}'(t) \cdot \vec{r}''(t) = t + 2(2t + 1) + 6t(3t^2 - 1) = 18t^3 - t + 2$$

$$\begin{aligned}
 \vec{r}'(t) \times \vec{r}''(t) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ t & 2t + 1 & 3t^2 - 1 \\ 1 & 2 & 6t \end{vmatrix} = \begin{vmatrix} \vec{i} & \vec{j} \\ t & 2t + 1 \\ 1 & 2 \end{vmatrix} \\
 &= (6t)(2t + 1) \vec{i} + (3t^2 - 1) \vec{j} + 2t\vec{k} - 6t^2\vec{j} - 2(3t^2 - 1) \vec{i} - (2t + 1)\vec{k} \\
 &= (6t^2 + 6t + 2) \vec{i} - (3t^2 + 1) \vec{j} - \vec{k}
 \end{aligned}$$

Next, we also need a couple of magnitudes.

$$\|\vec{r}'(t)\| = \sqrt{t^2 + (2t + 1)^2 + (3t^2 - 1)^2} = \sqrt{9t^4 - t^2 + 4t + 2}$$

$$\|\vec{r}'(t) \times \vec{r}''(t)\| = \sqrt{(6t^2 + 6t + 2)^2 + (3t^2 + 1)^2 + 1} = \sqrt{45t^4 + 72t^3 + 66t^2 + 2}$$

The tangential component of the acceleration is then,

$$a_T = \frac{18t^3 - t + 2}{\sqrt{9t^4 - t^2 + 4t + 2}}$$

The normal component of the acceleration is,

$$a_N = \frac{\sqrt{45t^4 + 72t^3 + 66t^2 + 24t + 6}}{\sqrt{9t^4 - t^2 + 4t + 2}} = \sqrt{\frac{45t^4 + 72t^3 + 66t^2 + 24t + 6}{9t^4 - t^2 + 4t + 2}}$$